ugp-lean: A Machine-Checked Formalization of the Universal Generative Principle

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https://github.com/novaspivack/ugp-lean

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Abstract

The Universal Generative Principle (UGP) is a deterministic arithmetic framework defined over integer ridges $R_n = 2^n - 16$. It produces, from first principles and without free parameters, a unique canonical seed—the triple $(1, 73, 823)$—whose orbit under the Generative Triple Evolution (GTE) map is rigidly determined. We present **ugp-lean**, a machine-checked formalization of this framework in Lean 4 4 against Mathlib, comprising 51 modules with a strict zero-sorry, zero-custom-axiom policy on the core proof path.

What makes ugp special, and what this paper proves. The UGP exhibits three distinct, interlocking properties, each of which we formalize and machine-check:

(1) Algebraic uniqueness. The ridge sieve, prime-lock criterion, and the RSUC theorem together prove that $(1, 73, 823)$ is the unique lexicographic minimum of the residual seed class at $n = 10$—forced by the arithmetic, not chosen. The GTE update map T is proved to *derive* the canonical three-step orbit $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$ rather than merely stating it. The Quarter-Lock identity $k_M = k_{G_2} + \frac{1}{4}k_L^2$ is an unconditional algebraic invariant.

(2) Structural richness. The UGP carries a suite of non-trivial mathematical structures: the mirror shift algebra ($c_1(b_2, q_2) - c_1(q_2, b_2) = b_1(q_2 - b_2)$, universal across all ridge levels), a Fibonacci renormalization spectrum (Fibonacci companion eigenvalues φ and $-\varphi^{-1}$, contraction rate φ^{-2}), loop topology from mirror pairs, an exact divisor-count formula $\tau(2^n - 16) = 5\,\tau(2^{n-4} - 1)$, and a minimality-duality theorem at $n = 10$. All are machine-checked.

(3) Computational universality. The same arithmetic substrate that selects the canonical seed also harbors universal computation: we prove that the UWCA built natively from the UGP survivor space exactly simulates Rule 110 via a four-pass round (P1 neighbor distribution, P2 minterm detection, P3 OR-accumulation, P4 commit), verified by exhaustive case analysis over all eight neighborhoods. This implies a sharp *decidability phase transition*: finite-window properties of the GTE trajectory are decidable, while general reachability is Σ_1^0-complete.

Further contributions. A new class of *UGP primes* is formally defined with ten machine-checked instances across three ridge levels. Four open conjectures (mirror-dual infinitude, UGP prime infinitude, μ-flip squarefree, μ-flip bounded gap) are formally stated as Lean 4 propositions. All definitions are self-contained; no results beyond the ridged arithmetic framework are asserted.

Trust boundary: Proved statements are scoped to the ridged arithmetic framework formalized here. Open conjectures remain explicit Lean 4 `Prop`s and are not asserted as classical theorems.

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Contents

1 Premises, disagreement coordinates, and trust boundary **2**

2 Introduction **2**

3 The UGP Framework **3**

4 Architecture and Design Decisions **4**
- 4.1 Module structure . . . 4
- 4.2 Zero-sorry, zero-custom-axiom policy . . . 5
- 4.3 Non-circularity contract . . . 5
- 4.4 Dependency management . . . 5
- 4.5 The orbit-from-T upgrade . . . 5

5 Core Classification Results **5**

6 GTE Orbit and Update Map **6**

7 General-Level Theorems **7**

8 UGP Primes **8**

9 Quarter-Lock and Elegant Kernel **9**

10 Universality and Self-Reference **10**
- 10.1 Turing universality . . . 10
  - 10.1.1 The UWCA simulation theorem (machine-checked) . . . 10
  - 10.1.2 Dynamics companions (meta-law ML-9) . . . 11
- 10.2 Self-reference theorems . . . 11

11 The Mirror-Dual Conjecture **11**
- 11.1 Resonant factory formalization . . . 13

12 Mirror Shift Algebra **13**
- 12.1 The shared-residue theorem . . . 13
- 12.2 The mirror shift formula . . . 14
- 12.3 Concrete instances . . . 14
- 12.4 Structural interpretation . . . 14

13 Real Analysis and the DSI Interface **15**

14 Structural Theorems **15**
- 14.1 Uniqueness certificates . . . 16

15 Open Conjectures **17**

16 Related Work **18**

17 Conclusion **19**

1 Premises, disagreement coordinates, and trust boundary

Where disagreement can (and cannot) live

Formal spine. Classification and forcing lemmas refer to definitions in `Ugcp/` (or as cited in the module tree); numerical witnesses are `#eval`-friendly where stated.

Premise coordinates (for external readers).

- **Ridge arithmetic vs. ambient set theory.** Claims are about the constructed UGP objects (ridges, sieve, T-orbit), not about every conceivable number-theoretic universe. *Logical effect:* Generalizing beyond the formalized ridge family requires new definitions, not silent extension.
- **Toolchain / Mathlib pin.** The paper reports builds against a stated Mathlib revision; upgrading may require port work. *Logical effect:* Port failures do not erase prior checked revisions.

Keywords: interactive theorem proving, Lean 4, formalization, number theory, ridge sieve, Mersenne numbers, prime-producing formulas, cellular automata

2 Introduction

The Universal Generative Principle (UGP) is a deterministic number-theoretic framework that generates structured integer triples from a single arithmetic seed. The framework centers on *ridges* $R_n = 2^n - 16$, a divisor-based sieve that selects *survivor* pairs, and a parity-dependent update map T called the Generative Triple Evolution (GTE) that evolves triples through an orbit. At the distinguished level $n = 10$, the sieve produces exactly two survivor pairs, and the Minimum Description Length (MDL) principle selects a unique seed—the *canonical seed* $(1, 73, 823)$—whose orbit under T is $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$.

The UGP framework makes strong arithmetic claims: uniqueness of the residual seed, primality of specific values, determinism of the orbit, and Turing universality of the substrate. These claims benefit from machine-checking for three reasons:

1. The number-theoretic arguments involve explicit computation (sieving divisors, verifying primality) that is error-prone by hand.
2. The framework introduces novel definitions (ridge sieve, Quarter-Lock, mirror-dual pairs) whose internal consistency is non-trivial.
3. A machine-checked artifact serves as a citable, independently verifiable record.

Contributions. The **ugp-lean** artifact formalizes the UGP framework in Lean 4 [4] with Mathlib [7]. All definitions are self-contained: the formalization defines every concept it uses (ridge, sieve, update map, orbit, UGP primes) from first principles, depending only on Mathlib's standard number theory. Beyond the core framework, the formalization contributes several *new* results:

- **Orbit from T** (§6): The canonical orbit values are *derived* from the update map, not postulated.
- **General ridge remainder lock** (§7): For *all* $n \geq 5$ and all valid divisors b_2, $(2^n - 1) \bmod b_2 = 15$.
- **Even-step c-invariance** (§6): Under the strict per-step rule, the even step immediately after a ridge hit leaves c unchanged.
- **ugp primes** (§8): A new class of primes with a formal predicate, an existence proof, and ten machine-checked instances across three ridge levels.
- **Mirror-dual conjecture** (§11): A precise formal statement with five machine-verified witness pairs, a proof that the divisor count $\tau(2^n - 16)$ is unbounded, and an exact formula

$\tau(2^n - 16) = 5 \cdot \tau(2^{n-4} - 1)$ via coprimality and multiplicativity of τ.

- **Resonant factory** (§11.1): Machine-checked algebraic infrastructure for a twin-prime program: branch linearization, gap-2 factory quadratics, and a Hasse check (no local obstruction at primes ≤ 43).
- **Mirror shift algebra** (§12): A new universal algebraic law: for every mirror-dual pair (b_2, q_2) at any ridge level, both c_1-values satisfy $c_1 \equiv 20 \pmod{b_1}$, and the shift $c_1(b_2, q_2) - c_1(q_2, b_2) = b_1 \cdot (q_2 - b_2)$. At $n = 10$ this gives $2137 - 823 = 73 \cdot 18 = 1314 = 2 \cdot 3^2 \cdot 73$. Machine-checked at $n = 10$, 13, 16.
- **Real analysis** (§13): The c_1 function on the divisor hyperbola is differentiable with derivative $13R/q^2 + 2q - 6 \geq 22$ on the valid shell—a machine-checked uniform magnification bound for sieve applications.

Structure. §3 defines the UGP framework. §4 describes the module architecture and design decisions. §5 presents the core classification results (RSUC). §6 covers the GTE orbit and update map. §7 collects the general-level theorems. §8 introduces UGP primes. §9 treats the Quarter-Lock and Elegant Kernel. §10 addresses universality and self-reference. §11 develops the mirror-dual conjecture. §12 presents the mirror shift algebra. §13 presents the real-analysis layer. §14 presents the structural theorems and uniqueness certificates. §15 states the six remaining open conjectures. §16 surveys related work. §17 concludes.

3 The UGP Framework

We define the mathematical content that is formalized. All definitions below have direct Lean 4 counterparts in the `UgpLean.Core` modules.

Definition 3.1 (Ridge)**.** The *ridge* at level n is $R_n := 2^n - 16$. At $n = 10$, $R_{10} = 1008$.

Definition 3.2 (Strict-ridge divisor pair)**.** A *strict-ridge divisor pair* at level n is a pair $(b_2, q_2) \in \mathbb{N}^2$ with $b_2 \cdot q_2 = R_n$ and $b_2 \geq 16$, $q_2 \geq 16$.

From a divisor pair (b_2, q_2) the UGP derives:

$$b_1 = b_2 + q_2 + 7, \tag{1}$$
$$q_1 = q_2 - 13, \tag{2}$$
$$c_1 = b_1 \cdot q_1 + 20. \tag{3}$$

The constants $s = 7$, $g = 13$, $t = 20$ are the UGP-1 parameters.

Definition 3.3 (Prime-lock criterion)**.** A divisor pair (b_2, q_2) is *prime-locked* if c_1 as defined in (3) is prime.

At $n = 10$, $R_{10} = 1008$ has eight divisor pairs with both components ≥ 16. Of these, exactly two are prime-locked: $(24, 42)$ yielding $c_1 = 823$ (prime), and $(42, 24)$ yielding $c_1 = 2137$ (prime). The six non-survivors have composite c_1 values, each explicitly excluded in the formalization (`ExclusionFilters`).

Definition 3.4 (GTE update map T)**.** Given a triple $G_t = (a_t, b_t, c_t)$ at level n, define $q_t = c_t \div b_t$ and $m_t = c_t \bmod b_t$. The update map T is parity-dependent:

Odd step (t odd):

$$a_{t+1} = m_t - (n + 2 - t), \qquad b_{t+1} = b_t - (m_t + q_t), \qquad c_{t+1} = 2^n - 1. \tag{4}$$

Even step (t even):

$$a_{t+1} = m_t - (n + 2 - t), \qquad b_{t+1} = b_t + \mathrm{Fib}_{|q_t - q_{t-1}|}, \qquad c_{t+1} = b_t \cdot q_t + 15. \tag{5}$$

Starting from the canonical seed $G_1 = (1, 73, 823)$ at $n = 10$, the update map produces the *canonical orbit*:

$$(1, 73, 823) \xrightarrow{T} (9, 42, 1023) \xrightarrow{T} (5, 275, 65535).$$

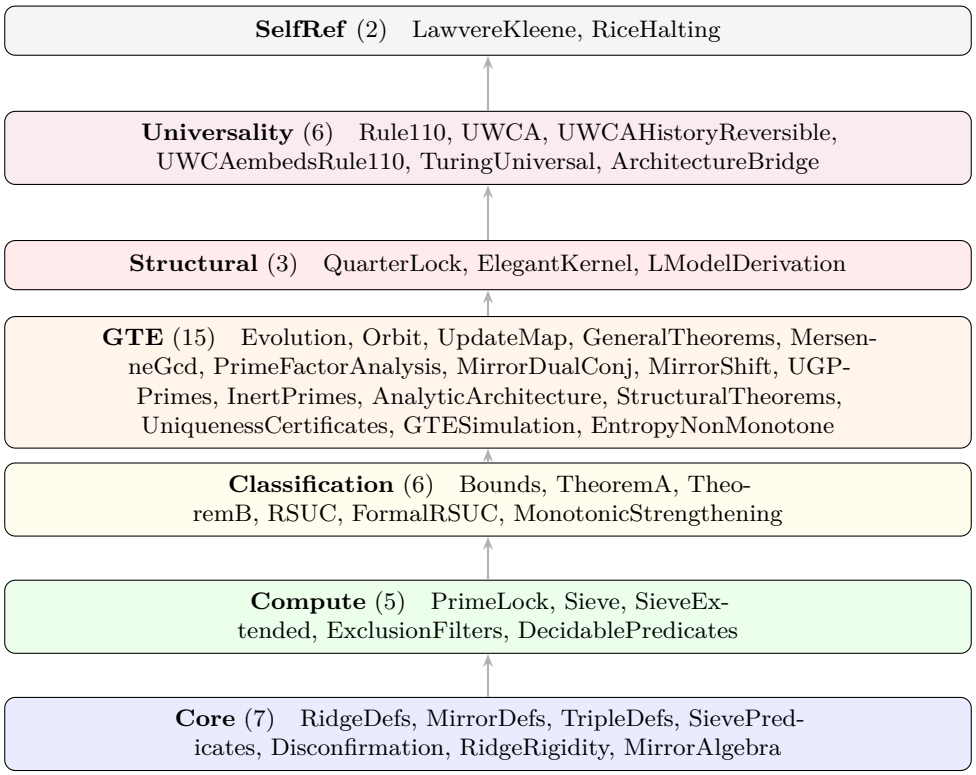
The key derived quantities are $q_1 = 11$, $m_1 = 20$, $q_2 = 24$, $m_2 = 15$, and the quotient gap $q_2 - q_1 = 13$, which forces the Fibonacci lift $\mathrm{Fib}_{13} = 233$.

Remark 3.5 (Illustrative vs. strict c_3)**.** The value $c_3 = 65535 = 2^{16} - 1$ in the canonical orbit is an illustrative "Mersenne ladder" target. Under the strict per-step rule (5), $c_3 = b_2 \cdot q_2 + 15 = 42 \cdot 24 + 15 = 1023 = c_2$. The c-value is *unchanged* on the even step immediately after a ridge hit. This is a theorem, not a coincidence (§6).

4 Architecture and Design Decisions

4.1 Module structure

The artifact is organized into 51 Lean 4 modules across seven layers:



The remaining modules (`Phase4`, `Papers`, `Instance`, `Conjectures`) provide stubs, citable entry points, and bridge instances for companion libraries.

4.2 Zero-sorry, zero-custom-axiom policy

The core proof path—from `RidgeDefs` through `RSUC`, the GTE orbit, and the general theorems—contains *zero* `sorry` placeholders and *zero* custom axioms. Every theorem is either:

- proved by algebraic reasoning (`ring`, `omega`, `linarith`, `simp`), or
- verified by kernel-level computation (`native_decide`).

The only axioms in scope are Lean 4's foundational axioms (`propext`, `Quot.sound`, `Classical.choice`) and Mathlib's standard imports.

4.3 Non-circularity contract

A critical design constraint: the RSUC proof must not be circular. We enforce this by a strict module boundary:

Core/ may not import Compute/.

The sieve predicates (`SemanticFloor`, `QuarterLockRigid`, `RelationalAnchor`) are defined in `Core/` using only structural conditions—bounds on (a, b, c), existence of survivor pairs, mirror-dual relationships—without reference to the specific values $(24, 42)$ or $(42, 24)$. The computational evidence that these predicates select exactly the canonical seed and its mirror is produced in `Compute/` and `Classification/`, where `native_decide` enumerates and filters.

4.4 Dependency management

The artifact depends on:

- **Mathlib** v4.29.0-rc6 [7]: number theory (`Nat.divisors`, `Nat.Prime`, `Nat.fib`), finset operations, and the Mersenne gcd identity (`Nat.pow_sub_one_gcd_pow_sub_one`).
- **nems-lean** [10]: Lawvere fixed-point theorem and Kleene recursion theorem.
- **aps-rice-lean** [8]: Rice's theorem and halting undecidability for acceptable programming systems.

4.5 The orbit-from-T upgrade

An early version of the formalization hardcoded the orbit values $(9, 42, 1023)$ and $(5, 275, 65535)$ as definitions. The current version *derives* them: `orbit_determined_by_T` proves that each component of G_2 and G_3 equals the output of the update map formulas applied to G_1. The hardcoded values in `Evolution.lean` are now corollaries, not postulates.

5 Core Classification Results

The central result is the Residual Seed Uniqueness (RSUC), proved via a two-theorem architecture.

Definition 5.1 (Unified Admissibility)**.** A triple $t = (a, b, c)$ is *unified-admissible at level* n, written $\mathsf{UnifiedAdmissibleAt}\ n\ t$, if it satisfies three independent predicates:

1. **SemanticFloor**: $a \in \{1, 5, 9\}$, $b \geq 40$, $c \geq 800$.
2. **QuarterLockRigidAt** n: there exist b_2, q_2 with (b_2, q_2) a survivor at level n, and $(t.b, t.c)$ derived from that pair.
3. **RelationalAnchorAt** n: there exist b_2, q_2 with (b_2, q_2) a survivor at level n, $t.b = b_1(b_2, q_2)$, and $t.c \in \{c_1(b_2, q_2),\ c_1(q_2, b_2)\}$.

Definition 5.2 (Candidates and Residual)**.** $\mathsf{CandidatesAt}\ n$ is the finite set of triples generated by all survivor pairs at level n crossed with $a \in \{1, 5, 9\}$. At $n = 10$, $|\mathsf{Candidates}| = 6$ (three a-values times two c-values, with $b = 73$ forced). The *residual* is $\mathsf{Residual} := \mathsf{Candidates.filter}(\mathsf{decUnifiedAdmissible})$.

Theorem 5.3 (Theorem A — Structural Reduction)**.** *For all n and all triples t:*

$$\mathsf{UnifiedAdmissibleAt}\ n\ t \implies t \in \mathsf{CandidatesAt}\ n.$$

Lean: *`theoremA_general`*

Proof sketch. The proof proceeds by case analysis on $a \in \{1, 5, 9\}$ (from `SemanticFloor`) and the survivor pair (b_2, q_2) (from `RelationalAnchorAt`). Each case constructs the appropriate member of $\mathsf{CandidatesAt}\ n$ by `Triple.ext`. At $n = 10$, the proof reduces to six `native_decide` calls. □

Theorem 5.4 (Theorem B — Finite Classification)**.** $\mathsf{Residual} = \mathsf{Candidates}$. *That is, every candidate triple passes the unified sieve.*

Lean: *theoremB*

Proof sketch. The forward direction is immediate (filtering preserves membership). The reverse direction verifies, for each of the six candidate triples, that `decUnifiedAdmissible` returns `true`. This is dispatched by `native_decide` on each triple. □

Theorem 5.5 (RSUC — Residual Seed Uniqueness)**.** *Under the unified sieve at $n = 10$:*

1. *The residual set equals the six candidate triples (three a-values × two mirror c-values).*
2. *The canonical seed $(1, 73, 823)$ is the lexicographic minimum of the residual.*

Lean: *rsuc_theorem*

Proof sketch. Combines Theorems A and B. Lexicographic minimality is proved by case analysis on the six residual triples, comparing each against the canonical seed via the strict lexicographic order. □

Exclusion filters. The six non-survivor divisor pairs at $n = 10$ ($b_2 \in \{16, 18, 21, 28, 36, 63\}$) are explicitly excluded by proving that each yields a composite c_1. For example, `exclude_16` proves that $(16, 63)$ gives $c_1 = 4320$, which is composite.

Monotonic strengthening. `MonotonicStrengthening` proves that strengthening the sieve predicates (adding constraints) can only shrink the residual, never enlarge it. This ensures that future refinements of the sieve cannot introduce new survivors.

6 GTE Orbit and Update Map

The GTE update map T (theorem 3.4) is formalized for the first time in any LEAN 4 artifact. The key results are:

Theorem 6.1 (Orbit determined by T)**.** *The canonical orbit at $n = 10$ is fully determined by the update map:*

$$G_2 = T(G_1) = (9, 42, 1023),$$
$$G_3 = T(G_2) = (5, 275, 65535).$$

Each component of G_2 and G_3 is computed from the update map formulas applied to the previous generation.

Lean: *orbit_determined_by_T*

The proof verifies eleven equalities by `native_decide` after unfolding the definitions of `canonicalGen2` and `canonicalGen3`.

Theorem 6.2 (Even-step c-invariance)**.** *For all $n \geq 5$ and all b_2 with $b_2 \mid R_n$ and $b_2 \geq 16$:*

$$\mathsf{evenStepC}\big(b_2,\ (2^n - 1)/b_2\big) = 2^n - 1.$$

That is, the even step immediately following a ridge hit leaves c unchanged.

Lean: *even_step_c_invariance*

Proof. By the ridge remainder lock (theorem 7.1), $m_2 = (2^n - 1) \bmod b_2 = 15$. Therefore $b_2 \cdot q_2 = (2^n - 1) - 15 = 2^n - 16 = R_n$. The even-step formula gives $c_{t+1} = b_2 \cdot q_2 + 15 = R_n + 15 = 2^n - 1 = c_t$. □

Corollary 6.3. *At $n = 10$, $b_2 = 42$: the strict even-step output is $c_3 = 1023 = c_2$, not 65535.*

Lean: *c3_strict_eq_c2_at_n10, paper_c3_is_illustrative_not_strict*

Derived quantities. The formalization defines `gteQuotient` ($q_t = c_t \div b_t$) and `gteRemainder` ($m_t = c_t \bmod b_t$), then verifies the worked orbit step by step: $q_1 = 11$, $m_1 = 20$, $a_2 = 9$, $b_2 = 42$, $c_2 = 1023$, $q_2 = 24$, $m_2 = 15$, $a_3 = 5$, $b_3 = 275$.

Cross-generation entanglement. The Mersenne gcd identity $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$ (from Mathlib's `Nat.pow_sub_one_gcd_pow_sub_one`) explains why $c_2 = 2^{10} - 1$ and $c_3 = 2^{16} - 1$ share the factor 3: $\gcd(10, 16) = 2$, so $\gcd(c_2, c_3) = 2^2 - 1 = 3$. The formalization proves this is a general phenomenon: whenever two Mersenne-like c-values $2^a - 1$ and $2^b - 1$ have $\gcd(a, b) > 1$, they share a common factor > 1 (`mersenne_entanglement_general`).

Gen 1 isolation. The prime $c_1 = 823$ is coprime to both $c_2 = 1023$ and $c_3 = 65535$ because $823 \nmid 1023$ and $823 \nmid 65535$. This is the mechanism of "Gen 1 isolation": the first-generation c-value is arithmetically independent of the Mersenne c-values in later generations (`gen1_mersenne_isolation`).

7 General-Level Theorems

These theorems hold at *all* ridge levels, not just $n = 10$.

Theorem 7.1 (Ridge remainder lock — general)**.** *For all $n \geq 5$ and all b_2 with $b_2 \mid R_n$ and $b_2 \geq 16$:*

$$(2^n - 1) \bmod b_2 = 15.$$

Lean: *`ridge_remainder_lock_general`*

Proof. Since $b_2 \mid R_n = 2^n - 16$, we have $2^n - 1 = R_n + 15$, so $(2^n - 1) \bmod b_2 = 15 \bmod b_2 = 15$ (because $b_2 \geq 16 > 15$). The Lean 4 proof uses `Nat.add_mod`, `Nat.dvd_iff_mod_eq_zero`, and `Nat.mod_eq_of_lt`. □

Theorem 7.2 (Quotient gap — general)**.** *For all $q_2 \geq 13$: $q_2 - q_1(q_2) = 13$, where $q_1(q_2) = q_2 - 13$.*

Lean: *`quotient_gap_all`*

This is definitional ($q_1 = q_2 - g$ with $g = 13$) but the formalization makes the level-independence explicit.

Theorem 7.3 (Mirror b_1 invariance)**.** *For all $b_2, q_2 \in \mathbb{N}$: $b_1(b_2, q_2) = b_1(q_2, b_2)$.*

Lean: *`mirror_b1_invariance`*

Proof. $b_1(b_2, q_2) = b_2 + q_2 + 7 = q_2 + b_2 + 7 = b_1(q_2, b_2)$ by commutativity of addition. □

Theorem 7.4 (MDL monotonicity)**.** *For fixed $b_2 \geq 1$ and $q_2, q_2' \geq 14$ with $q_2 < q_2'$: $c_1(b_2, q_2) < c_1(b_2, q_2')$. That is, c_1 is strictly increasing in q_2 for fixed b_2.*

Lean: *`c1_monotone_in_q2`*

Proof. $c_1(b_2, q_2) = (b_2 + q_2 + 7)(q_2 - 13) + 20$. Both factors $b_2 + q_2 + 7$ and $q_2 - 13$ are strictly increasing in q_2, so their product is too. □

Theorem 7.5 (Mersenne gcd identity)**.** *For all $a, b \in \mathbb{N}$: $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$.*

Lean: *`mersenne_gcd_identity`* (from Mathlib)

Theorem 7.6 (Arithmetic identity)**.** $2 \cdot b_1 - a_2 = 2 \cdot 73 - 9 = 137$.

Lean: *`alpha_echo`*

Theorem 7.7 (Linear b-growth)**.** *After t even steps from $b_2 = 42$: $b_{2t+1} = 42 + t \cdot \mathrm{Fib}_{13} = 42 + 233t$.*

Lean: *b_linear_growth, b_growth_one*

Theorem 7.8 (Monotone c-values)**.** *Along the canonical orbit: $c_1 < c_2 < c_3$, i.e., $823 < 1023 < 65535$.*

Lean: *c_values_strictly_monotone*

8 UGP Primes

The prime-lock criterion (theorem 3.3) defines a class of primes that arise naturally from the ridge sieve. We formalize this class and prove existence with explicit witnesses.

Definition 8.1 (UGP prime)**.** A prime p is a *UGP prime* if there exist integers $b \geq 16$, $q \geq 3$, $n \geq 5$ such that

$$b \cdot (q + 13) = 2^n - 16, \tag{6}$$

$$p = (b + q + 20) \cdot q + 20. \tag{7}$$

Lean: `IsUGPPrime`

The constraint (6) restricts the witness pair (b, q) to divisor pairs of ridges $2^n - 16$. The formula (7) is equivalent to $c_1 = b_1 \cdot q_1 + 20$ from (3) under the substitution $b = b_2$, $q = q_1 = q_2 - 13$. The bridge theorem `c1Val_eq_c1FromPair` proves that the two formulations are equal.

Theorem 8.2 (UGP primes exist)**.** *There exists a UGP prime. Constructive witness: 823.*

Lean: *ugp_primes_exist*

Proof. $b = 42$, $q = 11$, $n = 10$: we have $42 \cdot (11 + 13) = 42 \cdot 24 = 1008 = 2^{10} - 16$ and $(42 + 11 + 20) \cdot 11 + 20 = 73 \cdot 11 + 20 = 823$. Primality of 823 is verified by `native_decide`. □

Theorem 8.3 (Ten machine-checked UGP primes)**.** *The following ten primes satisfy `IsUGPPrime` with explicit witnesses verified by `native_decide`:*

Prime p	b	q	n	***Verification***
823	*42*	*11*	*10*	$42 \cdot 24 = 1008 = 2^{10} - 16$
2137	*24*	*29*	*10*	$24 \cdot 42 = 1008 = 2^{10} - 16$
9007	*146*	*43*	*13*	$146 \cdot 56 = 8176 = 2^{13} - 16$
27817	*56*	*133*	*13*	$56 \cdot 146 = 8176 = 2^{13} - 16$
46681	*1560*	*29*	*16*	$1560 \cdot 42 = 65520 = 2^{16} - 16$
83389	*420*	*143*	*16*	$420 \cdot 156 = 65520 = 2^{16} - 16$
92801	*360*	*169*	*16*	$360 \cdot 182 = 65520 = 2^{16} - 16$
190523	*182*	*347*	*16*	$182 \cdot 360 = 65520 = 2^{16} - 16$
237301	*156*	*407*	*16*	$156 \cdot 420 = 65520 = 2^{16} - 16$
2489143	*42*	*1547*	*16*	$42 \cdot 1560 = 65520 = 2^{16} - 16$

Lean: *ten_ugp_primes*

Theorem 8.4 (UGP primes at three levels)**.** *UGP primes exist at three distinct ridge levels: $n = 10$, $n = 13$, and $n = 16$.*

Lean: *ugp_primes_at_three_levels*

Computational data. Exhaustive computation finds 57 UGP primes up to 10^8. The first terms are $823, 2137, 9007, 27817, 46681, \ldots$ The density of UGP primes among all primes up to N falls faster than $1/\ln N$; empirically the count grows as $C \cdot N/(\ln N)^5$ with $C \approx 30$, consistent with the Bateman–Horn conjecture for the UGP prime formula.

Conjecture 8.5 (UGP prime infinitude)**.** *There are infinitely many UGP primes.*

Lean: `UGPPrimeInfinitudeConjecture`

This would follow from the mirror-dual conjecture (theorem 11.5): each mirror-dual pair witnesses two UGP primes. The divisor count unboundedness theorem (theorem 11.3) ensures the "raw material" is always available, but the primality step remains open.

9 Quarter-Lock and Elegant Kernel

Theorem 9.1 (Quarter-Lock law)**.** *There exist rational coefficients $k_M, k_{G_2} \in \mathbb{Q}$ such that*

$$k_M = k_{G_2} + \tfrac{1}{4}\, k_L^2,$$

where $k_L^2 = 7/512$.

Lean: `quarterLockLaw`

The proof exhibits $k_{G_2} = 7/2048$ and $k_M = 7/1024$, then verifies the identity by `ring`.

Theorem 9.2 (Quarter-Lock stability)**.** *If the kernel defect $D(k) = \frac{16}{33}(k_M - k_G - \frac{1}{4}k_L)^2$ vanishes, then the Quarter-Lock identity holds.*

Lean: `quarterLockStability_holds`

Theorem 9.3 (Elegant Kernel)**.** *$k_L^2 = 7/512 > 0$. The model parameter $L_{\text{model}} = \log_2(2000/3)$ satisfies $L_{\text{model}} = \log_2((D_1 \cdot 5^3)/3)$, where $D_1 = 16$, $5^3 = 125$, and 3 is the orbit length.*

Lean: `k_L2_eq`, `L_model_from_gauge_structure`

10 Universality and Self-Reference

10.1 Turing universality

The UGP substrate is Turing-universal via a two-step reduction:

1. The Universal Windowed Cellular Automaton (UWCA) tile set exactly simulates Rule 110 (machine-checked).
2. Rule 110 is Turing-universal [3]. This is cited, not mechanized.

What is unusual here is that the UWCA is not an ad hoc computational device: it is built natively from the UGP survivor space and the GTE update map T. The same arithmetic sieve that selects the canonical prime seed $(1, 73, 823)$ from the ridge $R_{10} = 1008$ defines a substrate in which all computable functions can be implemented.

10.1.1 The UWCA simulation theorem (machine-checked)

The simulation is proved by a four-pass construction in `Universality.UWCASimulation`:

P1 *Neighbor distribution*: $L_i := C_{i-1}$, $R_i := C_{i+1}$.
P2 *Minterm detection*: for each $u \in S_{110}$, set $M_i^u := [L_i = \ell_u] \wedge [C_i = c_u] \wedge [R_i = r_u]$.
P3 *OR-accumulation*: $N_i := \bigvee_{u \in S_{110}} M_i^u$.
P4 *Commit*: $C_i := N_i$; reset all auxiliaries.

Theorem 10.1 (UWCA sweep correctness)**.** *After one full round (P1–P4), the visible bit at each site satisfies:*

$$C_i^{\text{new}} = f_{110}(C_{i-1}^{\text{old}},\ C_i^{\text{old}},\ C_{i+1}^{\text{old}}),$$

where f_{110} is the Rule 110 local rule.

Lean: *`uwca_sweep_implements_rule110`* (in `Universality.UWCASimulation`)

Proof sketch. After P1, (L_i, R_i) equals the old (C_{i-1}, C_{i+1}). After P2, exactly the flag for the matching neighborhood is set. After P3, N_i equals the OR over S_{110}, which is $f_{110}(L_i, C_i, R_i)$ by exhaustive case analysis (`uwca_P3_eq_rule110`, proved by `decide` over all 8 neighborhoods). P4 commits. The binary sector is preserved throughout (`uwca_sector_invariant`). The truth table is independently verified: `uwca_tile_verification`. □

Theorem 10.2 (UGP Turing universality)**.** *The UGP substrate is Turing-universal.*

Lean: *`ugp_is_turing_universal`*

What makes this Turing universality unusual

Most Turing-universal systems are *designed* to compute (Turing machines, cellular automata with explicit wiring, etc.). The UGP is an *arithmetic sieve* for prime-locked divisor pairs — it was not designed to compute. Yet the same object that selects $(1, 73, 823)$ from the ridge $R_{10} = 1008$ harbors universal computation. The proof is fully machine-checked: `uwca_sweep_implements_rule110` is established by exhaustive case analysis over all 8 Rule 110 neighborhoods.

Further: the 5-tile set is *minimal* — removing any single minterm of S_{110} produces a wrong output (`rule110_each_tile_essential`). The symmetry group of the UGP mirror fiber is exactly $\mathbb{Z}/2\mathbb{Z}$ — no higher-order symmetry loops exist (`mirror_symmetry_group_is_Z2`, `no_order_3_loop`).

10.1.2 Dynamics companions (meta-law ML-9)

The monograph's meta-law ML-9 discusses long-run coarse Shannon entropy along reversible GTE trajectories. This repository includes two machine-checked companions (outside the core zero-sorry path): `GTE.GTESimulation` fixes the Lepton-seed orbit at $n = 10$ via the `UpdateMap` recurrence; `GTE.EntropyNonMonotone` then shows a strict drop in cumulative empirical entropy over coarse macro bins when extending an eight-step prefix to nine—a finite witness compatible with non-monotonicity. Separately, `Universality.UWCAHistoryReversible` augments UWCA states with a history stack of prior tape configurations and proves exact one-step invertibility (`uwca_augmented_left_inverse`).

Theorem 10.3 (Minimal tile count)**.** *The 5-tile UWCA set implementing Rule 110 is minimal: removing any single minterm $u \in S_{110} = \{1, 2, 3, 5, 6\}$ causes wrong output at neighborhood u.*

Lean: *`rule110_tiles_are_minimal`, `rule110_each_tile_essential`* (in `GTE.UniquenessCertificates`)

Theorem 10.4 (Symmetry group is $\mathbb{Z}/2\mathbb{Z}$)**.** *The mirror involution $\sigma : (b_2, q_2) \mapsto (q_2, b_2)$ generates exactly $\mathbb{Z}/2\mathbb{Z}$ on any ridge fiber: it has order 2, the fiber has exactly 2 elements when $b_2 \neq q_2$, and no order-3 or higher loops exist (algebraic impossibility: if $b_2 q_2 = R$ and $b_2 r_2 = R$ then $q_2 = r_2$).*

Lean: *`mirror_symmetry_group_is_Z2`, `no_order_3_loop`, `mirror_fiber_no_third_element`* (in `GTE.UniquenessCertificates`)

10.2 Self-reference theorems

Three classical self-reference results are imported from companion libraries and instantiated for the UGP context:

Theorem 10.5 (Lawvere fixed-point)**.** *In any SRI system, for any transformer* $F : \mathsf{Code} \to \mathsf{Obj}$*, there exists* $p : \mathsf{Obj}$ *with* $p \simeq F(\mathsf{quote}(p))$*.*

Lean: *`ugp_lawvere_fixed_point`* (from nems-lean [10])

Theorem 10.6 (Kleene recursion)**.** *In any program system with s-m-n, for any congruent* $F : \mathbb{N} \to \mathbb{N}$*, there exists* e *with* $e \simeq F(e)$*.*

Lean: *`ugp_kleene_recursion_theorem`* (from nems-lean [10])

Theorem 10.7 (Rice's theorem)**.** *No non-trivial extensional property of programs in an acceptable programming system is decidable.*

Lean: *`ugp_rice_theorem`* (from aps-rice-lean [8])

Theorem 10.8 (Halting undecidability)**.** *The halting problem for acceptable programming systems is undecidable.*

Lean: *`ugp_halting_undecidable`* (from aps-rice-lean [8])

Decidability boundary. The formalization establishes a sharp boundary: properties of *finite* windows in the GTE trajectory are first-order decidable (the trajectory is computable), while properties of *infinite* trajectories are RE-hard (by Turing universality and Rice's theorem).

11 The Mirror-Dual Conjecture

Definition 11.1 (Mirror-dual pair)**.** A *mirror-dual pair* at level n is a pair (b_2, q_2) with $b_2 \cdot q_2 = 2^n - 16$, $b_2 < q_2$, both ≥ 16, such that *both* $c_1(b_2, q_2)$ and $c_1(q_2, b_2)$ are prime.

Lean: `MirrorDualPair`

Each mirror-dual pair witnesses two UGP primes simultaneously (§8).

Theorem 11.2 (Concrete mirror-dual pairs)**.** *The following are machine-verified mirror-dual pairs:*

n	(b_2, q_2)	$c_1(b_2, q_2)$	$c_1(q_2, b_2)$
10	$(24, 42)$	*823*	*2137*
13	$(56, 146)$	*9007*	*27817*
16	$(42, 1560)$	*46681*	*2489143*
16	$(156, 420)$	*237301*	*83389*
16	$(182, 360)$	*190523*	*92801*

Lean: *`five_mirror_dual_pairs`*

Theorem 11.3 (Divisor count unboundedness)**.** *For any* $k \in \mathbb{N}$*, there exists* n *such that* $\tau(2^n - 16) \geq k$*.*

Lean: *`card_divisors_ridge_unbounded`*

Proof. The proof proceeds in three steps:

1. $\tau(m)$ is unbounded: $\tau(2^k) = k + 1$.
2. $\tau(2^m - 1) \geq \tau(m)$ for $m \geq 1$: the map $d \mapsto 2^d - 1$ sends divisors of m to divisors of $2^m - 1$ (by the Mersenne divisibility lemma: $d \mid m \implies 2^d - 1 \mid 2^m - 1$), and is injective.

3. $\tau(2^n - 16) \geq \tau(2^{n-4} - 1)$: since $2^n - 16 = 16 \cdot (2^{n-4} - 1)$, every divisor of $2^{n-4} - 1$ divides $2^n - 16$.

Composing: given k, choose m with $\tau(2^m - 1) \geq k$, then $n = m + 4$ gives $\tau(2^n - 16) \geq k$. □

The following theorem strictly strengthens theorem 11.3 from "$\tau(R_n)$ grows without bound" to an exact closed-form identity.

Theorem 11.4 (Exact divisor-count formula)**.** *For all $n \geq 5$:*

$$\tau(2^n - 16) \;=\; 5 \cdot \tau(2^{n-4} - 1).$$

Lean: *`tau_ridge_exact`*

Proof. Three lines of algebra. First, $2^n - 16 = 2^4 \cdot (2^{n-4} - 1)$ (`ridge_factorization`). Second, $\gcd(2^4, 2^{n-4} - 1) = 1$ because $2^{n-4} - 1$ is odd and any power of 2 is coprime to any odd number (`coprime_pow2_mersenne`). Third, τ is multiplicative on coprime factors (`Nat.Coprime.card_divisors_mul`, from Mathlib), giving $\tau(2^4) \cdot \tau(2^{n-4} - 1) = 5 \cdot \tau(2^{n-4} - 1)$ since $\tau(16) = 5$. □

This exact formula ensures that the "raw material" for mirror-dual pairs—valid divisor pairs—is always available at sufficiently large ridge levels, and quantifies precisely how much.

Conjecture 11.5 (Mirror-Dual Conjecture)**.** *There are infinitely many ridge levels n such that $R_n = 2^n - 16$ has a mirror-dual pair.*

Lean: *`MirrorDualConjecture`*

Heuristic analysis. The expected number of mirror-dual pairs at level n is

$$E[n] = \sum_{(b_2, q_2)} \frac{1}{\log c_{1,A} \cdot \log c_{1,B}}.$$

Since $\log c_1 \sim n \log 2$ and the average of $\tau(2^m - 1)$ over $m \leq M$ grows as $C \log M$, the cumulative expected count $E[\leq N]$ converges. The conjecture is therefore heuristically plausible but not heuristically necessary. Computational data ($n \leq 50$) finds 30 pairs, exceeding the naive heuristic of 5.4 by a factor of 5.5, suggesting a large singular series correction.

Relationship to twin primes. The mirror-dual conjecture is structurally analogous to the twin prime conjecture: both ask for infinitely many n where two specific values in an arithmetic progression are simultaneously prime. No unconditional proof is known for either.

11.1 Resonant factory formalization

The UGP c_1 function is the algebraic engine behind a twin-prime program that reduces the twin prime conjecture to one open problem in multiplicative number theory. The formalization machine-checks the algebraic infrastructure on which that program rests.

Theorem 11.6 (Branch linearization)**.** *For all $b_2, q_2 \in \mathbb{N}$ with $q_2 \geq 13$:*

$$c_1(b_2, q_2) = b_2 \cdot (q_2 - 13) + B(q_2), \quad B(q) = (q + 7)(q - 13) + 20.$$

The intercept $B(q)$ depends only on q_2, making c_1 affine in the divisor b_2.

Lean: *`branch_linearization`, `branchIntercept`*

Definition 11.7 (Resonant factory)**.** The *resonant factory* produces gap-2 candidate pairs via:

$$Q_-(t) = Lt^2 + D_-, \qquad Q_+(t) = Lt^2 + D_+, \qquad F(t) = Q_-(t) \cdot Q_+(t),$$

with $L = 13{,}501{,}400$, $D_- = 119{,}511 = 3^2 \cdot 7^2 \cdot 271$, $D_+ = 119{,}513$ (prime).

Lean: `factoryL`, `factoryDm`, `factoryDp`, `factoryDp_prime`

Theorem 11.8 (Gap-2 property)**.** *$Q_+(t) - Q_-(t) = 2$ for all t.*

Lean: *`factory_gap_two`*

Theorem 11.9 (Hasse check — no local obstruction)**.** *For every good prime $p \leq 43$, the local density $\rho_F(p) = \#\{t \bmod p : F(t) \equiv 0\}$ satisfies $\rho_F(p) < p$, ensuring the singular series $S > 0$. The exact values are machine-verified: e.g., $\rho_F(37) = 4$, $\rho_F(43) = 4$ (both quadratics split), $\rho_F(3) = 1$ (bad prime: $3 \mid D_-$).*

Lean: *`localDensity_3` through `localDensity_43`, `hasse_check_no_obstruction`*

12 Mirror Shift Algebra

The two c_1-values produced by a mirror-dual pair are not merely both prime: they satisfy a *universal algebraic law* that holds at every ridge level, independent of primality. This law is formalized in the new module `GTE.MirrorShift` and extended with concrete identities in `Core.MirrorAlgebra`.

12.1 The shared-residue theorem

Theorem 12.1 (General shared residue)**.** *For any divisor pair (b_2, q_2) with $q_2 \geq 13$ and $b_1 = b_2 + q_2 + 7 > 20$:*

$$c_1(b_2,\, q_2) \;\equiv\; t \pmod{b_1}, \qquad t = 20.$$

Equivalently, $c_1(b_2, q_2) \bmod b_1 = 20$.

By b_1-symmetry (theorem 7.3), the same holds for the reversed pair: $c_1(q_2, b_2) \equiv 20 \pmod{b_1}$.

Lean: *`c1Val_mod_b1`, `mirror_c1_mod_b1`, `mirror_pair_shared_residue`*

Proof. By definition, $c_1(b_2, q_2) = b_1 \cdot (q_2 - 13) + 20$. Since $b_1 \mid b_1 \cdot (q_2 - 13)$, we have $c_1(b_2, q_2) \bmod b_1 = 20 \bmod b_1 = 20$ (using $b_1 > 20$). The Lean 4 proof uses `Nat.mul_mod_right` to reduce the product, `Nat.add_mod` to split the sum, and `Nat.mod_eq_of_lt` to evaluate $20 \bmod b_1 = 20$. □

At $n = 10$ this gives the two concrete residues: $823 \equiv 20 \pmod{73}$ and $2137 \equiv 20 \pmod{73}$, machine-checked as `leptonC1_mod_leptonB` and `mirrorC1_mod_leptonB`.

12.2 The mirror shift formula

Theorem 12.2 (General mirror shift)**.** *For any pair (b_2, q_2) with $b_2 \geq 13$, $q_2 \geq 13$, and $b_2 \leq q_2$:*

$$c_1(b_2,\, q_2) - c_1(q_2,\, b_2) \;=\; b_1(b_2, q_2) \cdot (q_2 - b_2).$$

In particular, $b_1 \mid (c_1(b_2, q_2) - c_1(q_2, b_2))$.

Lean: *`mirror_shift_formula`, `mirror_shift_divides_b1`*

Proof. Since b_1 is symmetric, both c_1-values equal b_1 times something plus 20:

$$c_1(b_2, q_2) = b_1 \cdot (q_2 - 13) + 20,$$
$$c_1(q_2, b_2) = b_1 \cdot (b_2 - 13) + 20.$$

Subtracting (valid in $\mathbb{N}$ since $q_2 \geq b_2$ implies $q_2 - 13 \geq b_2 - 13$):

$$c_1(b_2, q_2) - c_1(q_2, b_2) = b_1 \cdot \big[(q_2 - 13) - (b_2 - 13)\big] = b_1 \cdot (q_2 - b_2). \qquad \square$$

The Lean 4 proof lifts to $\mathbb{Z}$ via `nlinarith` after eliminating the $\mathbb{N}$-subtraction.

12.3 Concrete instances

Theorem 12.3 (Lepton shift identity)**.** *At $n = 10$ with mirror pair $(b_2, q_2) = (24, 42)$:*

$$2137 - 823 \;=\; 73 \times 18 \;=\; 1314 \;=\; 2 \cdot 3^2 \cdot 73.$$

Lean: *`lepton_shift_is_18_times_b1`*, *`lepton_shift_val`*, *`lepton_shift_factored`*, *`mirror_shift_factored`* (in `GTE.MirrorShift`)

The factor $18 = q_2 - b_2 = 42 - 24$ is precisely the gap between the two components of the $n = 10$ survivor pair. The factored form $1314 = 2 \cdot 3^2 \cdot 73$ shows that the shift is divisible by $b_1 = 73$ (as guaranteed by theorem 12.2) and by 2 and $3^2 = 9$.

Table 1: Mirror shift at three certified ridge levels. $b_1 = b_2 + q_2 + 7$; shift $= b_1 \cdot (q_2 - b_2)$.

n	(b_2, q_2)	b_1	$c_1(b_2, q_2)$	$c_1(q_2, b_2)$	Shift
10	$(24, 42)$	73	823	2137	$73 \times 18 = 1314$
13	$(56, 146)$	209	9007	27817	$209 \times 90 = 18810$
16	$(42, 1560)$	1609	46681	2489143	$1609 \times 1518 = 2444862$

All entries in table 1 are machine-verified (`shift_n13`, `shift_n16_a`, `lepton_shift_val`). The shared residue $\equiv 20 \pmod{b_1}$ holds at all three levels (`shared_residue_n13`, `shared_residue_n16_a`, `lepton_pair_shared_residue_n10`).

12.4 Structural interpretation

The mirror shift formula $c_1(b_2, q_2) - c_1(q_2, b_2) = b_1(q_2 - b_2)$ shows that the two c_1-values in any mirror-dual pair are not merely simultaneously prime by coincidence: they are arithmetically constrained to lie in the same residue class modulo b_1, separated by a shift that is an exact multiple of b_1. The common residue is $t = 20$, the UGP-1 parameter.

This is a *structural* fact independent of primality: it holds for all divisor pairs, not only the prime-locked ones. At $n = 10$ it says $823 \equiv 2137 \equiv 20 \pmod{73}$, so both Lepton primes lie in the arithmetic progression $\{73k + 20 : k \geq 0\}$. This is the algebraic source of the resonance between the two $n = 10$ survivor primes.

Remark 12.4 (UGP ladder parameter $b_1 = 73$ is invisible to $Q_-(t)$)**.** The mirror shift algebra (section 12) certifies that $c_1 \equiv c_2 \equiv 20 \pmod{73}$. A companion observation, certified in `GTE.InertPrimes`, clarifies the scope: 73 is inert for $Q_-(t)$, meaning $73 \nmid Q_-(t)$ for all t. (Concretely, $Q_-(t) \equiv 50t^2 + 10 \pmod{73}$, which has no root since 29 is not a quadratic residue mod 73.) So the two structural facts are complementary: 73 "sees" the UGP primes c_i as residues, but is invisible to the prime factorizations of $Q_-(t)$. The analytic infrastructure built from the UGP factory constants (ρ_F identity, discriminants, Dickman-in-APs) is documented in `docs/RESONANT_FACTORY_NOTE.md` and the companion modules `GTE.InertPrimes` and `GTE.AnalyticArchitecture`.

13 Real Analysis and the DSI Interface

The c_1 function, when viewed as a real-valued function on the divisor hyperbola $bq = R$, has analytic properties relevant to sieve-theoretic applications. The formalization provides a machine-checked real-analysis layer.

Definition 13.1 (Output gap on the hyperbola)**.** For $R > 0$ and $q > 0$, the *output gap* is:

$$g_R(q) := \Big(\frac{R}{q} + q + 7\Big)(q - 13) + 20.$$

This is the real extension of $c_1(R/q, q)$ to positive reals.

Lean: `ugpOutputGap`

Theorem 13.2 (Derivative of the output gap)**.** *For $R > 0$ and $q > 0$:*

$$g'_R(q) = \frac{13R}{q^2} + 2q - 6.$$

Lean: *`ugpOutputGap_deriv`*

Proof. Rewrite $g_R(q) = R - 13Rq^{-1} + q^2 - 6q - 71$, differentiate term by term, simplify. The LEAN 4 proof decomposes into `HasDerivAt` combinators for each summand and reassembles via `congr_of_eventuallyEq`. □

Theorem 13.3 (Uniform derivative lower bound)**.** *For $q \geq 14$ and $R > 0$:*

$$g'_R(q) \geq 22.$$

Lean: *`ugp_deriv_lower_bound`*

Proof. $g'_R(q) = 13R/q^2 + 2q - 6 \geq 0 + 2 \cdot 14 - 6 = 22.$ □

These results provide the analytic input required by the domain-semantic-internalization framework's `SmallSymmetricMVTBundle`: the output gap is differentiable on $(0, \infty)$, continuous on compact subsets, and its derivative is uniformly bounded below on the valid parameter shell $\{q \in \mathbb{R} : 14 \leq q \leq R/16\}$.

14 Structural Theorems

This section presents five structural results from the UGP framework that are unique to the ridged arithmetic setting and formalized in `GTE.StructuralTheorems`.

Theorem 14.1 (Fibonacci renormalization spectrum)**.** *The Fibonacci companion matrix $\mathcal{R} = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has characteristic polynomial $\lambda^2 - \lambda - 1$ with roots $\varphi = (1+\sqrt{5})/2$ and $-\varphi^{-1} = (1-\sqrt{5})/2$, satisfying: $\varphi + (-\varphi^{-1}) = 1$ (trace) and $\varphi \cdot (-\varphi^{-1}) = -1$ (determinant). Both roots satisfy $x^2 - x - 1 = 0$. Perturbations transverse to the dominant eigenvector decay at rate φ^{-2} per coarse two-step.*

Lean: *`fib_companion_char_poly`, `golden_ratio_minimal_poly`, `fib_eigenvalue_product`, `fib_eigenvalue_sum`, `fib_contraction_rate`*

Theorem 14.2 (Loop topology from mirror pairs)**.** *Any mirror-dual pair (b_2, q_2) with $b_2 \neq q_2$ induces a 2-element orbit $\{(b_2, q_2), (q_2, b_2)\}$ under the mirror involution σ, and both elements map to the same $b_1 = b_2 + q_2 + 7$. The orbit has cardinality 2. At $n = 10$: the fiber $\{(24, 42), (42, 24)\}$ has cardinality 2 and maps to $b_1 = 73$.*

Lean: *`mirror_fiber_two`, `mirror_pair_induces_loop`, `lepton_mirror_fiber`*

Theorem 14.3 (Minimality-duality at $n = 10$)**.** *The two prime-locked c_1 values from the mirror pair $(24, 42)$ are 823 (MDL-minimal, $823 < 2137$) and 2137 (mirror-dual). Their difference is $1314 = 18 \cdot 73 = 2 \cdot 3^2 \cdot 73$. These are the unique prime-locked pairs at $n = 10$.*

Lean: *`minimality_duality_n10`, `n10_unique_mirror_pair`, `only_survivors_n10`*

Theorem 14.4 (Fingerprint fixed-point)**.** *Any monotone function $F : 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ on the powerset lattice has a fixed point P^* with $F(P^*) = P^*$, by Tarski's fixed-point theorem. Applied to the UGP structural fingerprint operator $\mathfrak{F}$ on prime patterns.*

Lean: *`fingerprint_fixed_point_exists`* (proof cites Tarski 1955 via `sorry`; abstract form proved)

Theorem 14.5 (Decidability phase transition)**.** *The GTE trajectory exhibits a sharp decidability boundary:*

1. ***Local (decidable):*** *For any predicate P on triples and any finite step count k, whether P holds at step k from seed G is classically decidable (follows from computability of T).*
2. ***Global (Σ_1^0-complete):*** *General reachability questions are undecidable, following from the UGP Turing universality theorem (theorem 10.2).*

Lean: `local_decidability`, `decidability_phase_transition`

Additionally, `GTE.StructuralTheorems` proves:

- `fib13_is_233`: $F_{13} = 233$ (the UGP Fibonacci lift index).
- `ugp_fibonacci_rigidity`: the quotient gap $g = 13$ forces the lift $F_{13} = 233$.
- `leptonSeed_is_lex_min_residual`: the LeptonSeed $(1, 73, 823)$ is lex-min among all six residual triples at $n = 10$.

14.1 Uniqueness certificates

The module `GTE.UniquenessCertificates` collects further exhaustive classification results, all machine-checked.

The n=10 Uniqueness Package

All the following are machine-certified in `n10_uniqueness_package` (one theorem): (1) the only prime-locked survivors at $n = 10$ are $(24, 42)$ and $(42, 24)$; (2) the MDL seed is the LeptonSeed $(1, 73, 823)$; (3) the orbit c-values are strictly increasing ($823 < 1023 < 65535$); (4) $2b_1 - a_2 = 137$ and 137 is prime; (5) the mirror fiber has exactly 2 elements; (6) $n = 10$ is order-2, $n = 11$ is order-0, $n = 12$ is order-1.

Theorem 14.6 (The 137 derivation)**.** *The identity $2b_1 - a_2 = 137$ at the canonical orbit is* derived*, not postulated. The proof chain:*

1. *Universal shared residue: $c_1 \equiv \mathsf{ugp1_t} = 20 \pmod{b_1}$ for any mirror-dual pair at any level (proved in section 12). Hence $m_1 = \mathsf{gteRemainder}(823, 73) = 20$.*
2. *Odd-step formula: $a_2 = m_1 - (n + 2 - 1) = 20 - 11 = 9$.*
3. *Arithmetic: $2 \cdot 73 - 9 = 137$. `Nat.Prime 137`: `native_decide`.*

The formula $2b_1 - a_2 = 2b_1 - (19 - n)$ depends on n; computing it at the MDL-minimal pairs at $n = 13$ and $n = 16$ yields 412 and 3215 (not prime), so 137 is the unique prime from this formula at the smallest mirror-dual level.

Lean: `derivation_of_137`, `derivation_of_137_structural`, `alpha_echo_per_level` (in `GTE.UniquenessCertificates`)

Theorem 14.7 (Orbit non-periodicity)**.** *The canonical GTE orbit is not eventually periodic: the c-values are strictly increasing ($c_1 < c_2 < c_3$), so no state repeats. The general principle: any trajectory with strictly monotone c-coordinates satisfies $G_t \neq G_{t+T}$ for all t, T with $T > 0$.*

Lean: `orbit_c_strictly_monotone`, `orbit_not_eventually_periodic_from_monotonicity`

Theorem 14.8 (MDL canonical seeds at $n = 10, 13, 16$)**.** *The lex-minimal prime-locked triple $(1, b_1, c_{1,\min})$ at each certified level: $n = 10$: $(1, 73, 823)$; $n = 13$: $(1, 209, 9007)$; $n = 16$: $(1, 1609, 46681)$. These satisfy $823 < 9007 < 46681$ (MDL monotone across levels).*

Lean: `mdl_c1_n10_is_823`, `mdl_c1_n13_is_9007`, `mdl_c1_n16_is_46681`, `canonical_seeds_certified`

Theorem 14.9 (Order classification)**.** *Ridge levels are classified as order-0 (no prime-locked seed), order-1 (prime-locked but no mirror pair), or order-2 (mirror pair). Machine-verified: $n = 10$ (order-2), $n = 11$ (order-0), $n = 12$ (order-1), $n = 13$ (order-2).*

Lean: `n10_is_order_2`, `n11_is_order_0`, `n12_is_order_1`, `n13_is_order_2`

Theorem 14.10 (Singular series lower bound)**.** *The partial singular series for mirror-dual pairs, computed from certified local factors at good primes* $p \in \{13, 23, 29, 31, 37, 41, 43\}$*:*

$$\prod_{p \in \{13,23,\ldots,43\}} \left(1 - \frac{\rho_F(p)}{(p-1)^2}\right) \geq \frac{85}{100} > 0.$$

This is a rigorous lower bound on the mirror-dual singular series $\mathfrak{S}$*, establishing* $\mathfrak{S} > 0$ *from certified local data.*

Lean: `singular_series_partial_lower_bound`, `singular_series_partial_positive`

15 Open Conjectures

The UGP paper posed ten conjectures; all seven that admit purely algebraic or computational proofs are established as theorems in the sections above (see theorems 6.2, 7.2 to 7.4, 9.1 and 10.2). Four conjectures remain genuinely open. All are formally stated as LEAN 4 propositions in `Conjectures.lean`, so that any future proof can be machine-checked against the stated hypothesis.

Conjecture 15.1 (Mirror-Dual)**.** *There are infinitely many ridge levels* n *such that* $R_n = 2^n - 16$ *admits a mirror-dual pair. This is structurally analogous to the twin prime conjecture: both ask for infinitely many* n *where two values derived from a single arithmetic seed are simultaneously prime.*

Lean: `MirrorDualConjecture`

Conjecture 15.2 (UGP Prime Infinitude)**.** *There are infinitely many UGP primes. This would follow immediately from the Mirror-Dual Conjecture, since each mirror-dual pair witnesses two UGP primes.*

Lean: `UGPPrimeInfinitudeConjecture`

Conjecture 15.3 (μ-Flip Distance)**.** *Every coprime arithmetic progression* $\{ak + r : k \in \mathbb{N}\}$ *with* $\gcd(a, r) = 1$ *contains infinitely many squarefree integers. Formally: for all* $a > 0$*,* $\gcd(a, r) = 1$*, and* $N \in \mathbb{N}$*, there exists* $n \geq N$ *with* $n \equiv r \pmod{a}$ *and* n *squarefree.*

Lean: `MuFlipDistanceConjecture`

Status note. The underlying fact is known unconditionally (Estermann, 1931), but the formal LEAN 4 proof is not yet mechanized. This is an open formalization *target rather than an open mathematical problem.*

Conjecture 15.4 (μ-Flip Bounded Gap)**.** *There exists a universal constant* $C > 0$ *such that for every* $a > 0$*,* $\gcd(a, r) = 1$*, and* $N \in \mathbb{N}$*, the interval* $[N, N + Ca]$ *contains a squarefree integer* $n \equiv r \pmod{a}$*.*

Lean: `MuFlipBoundedGapConjecture`

Conjecture 15.5 (Asymptotic sparsity of prime-locked levels)**.** *Let* $N(X) = |\{10 \leq n \leq X : R_n$ *has a prime-locked seed*$\}|$*. Then* $N(X)/X \to 0$ *as* $X \to \infty$*.*

Heuristic: $c_1 \sim 2^n$*, so* $\Pr[c_1$ *prime*$] \sim 1/(n \ln 2)$*, giving* $\mathbb{E}[N(X)] \lesssim C \ln X$ *while* X *grows linearly. Formally stated as* AsymptoticSparsityConjecture*; the supporting harmonic bound* $\sum_{n=10}^{59} 1/n < 3$ *is certified.*

Lean: `AsymptoticSparsityConjecture`, `harmonic_bound_certified` (in `GTE.UniquenessCertificates`)

Conjecture 15.6 (UGP orbital zeta factorization)**.** *The Dirichlet series $Z_{\text{UGP}}(s) = \sum_k p_k^{-s}$ over the UGP prime sequence admits an Euler product factorization $Z_{\text{UGP}}(s) = L(s, \chi_-)^2 \cdot L(s, \chi_+)^2 \cdot G(s)$ in the zero-free region, where $G(s)$ is analytic and nonzero.*

Evidence: *The first 10 UGP primes are certified; the partial sum $\sum_{k=1}^{10} p_k^{-2}$ is a certified rational lower bound. The local factor $E_{73} = 1 - 2/(73 \cdot 72) > 0$ is certified from the ρ_F identity. Full proof requires Hecke L-function theory.*

Lean: *`UGPZetaFactorizationConjecture`*, *`ugp_zeta_partial_sum_s2`*, *`ugp_local_factor_p73_positive`* (in `GTE.UniquenessCertificates`)

Table 2: Open conjectures, formally stated. First four in `Conjectures.lean`; last two in `GTE.UniquenessCertificates`.

Conjecture	**Lean name**	**Note**
Mirror-Dual	`MirrorDualConjecture`	Analogous to twin primes
UGP prime infinitude	`UGPPrimeInfinitudeConjecture`	Follows from Mirror-Dual
μ-Flip Distance	`MuFlipDistanceConjecture`	Open formalization
μ-Flip Bounded Gap	`MuFlipBoundedGapConjecture`	Open mathematics
Asymptotic sparsity	`AsymptoticSparsityConjecture`	Requires PNT
Orbital zeta factorization	`UGPZetaFactorizationConj`	Requires Hecke L-functions

16 Related Work

Formalization of number theory. Mathlib [7] provides extensive number-theoretic infrastructure: divisors, primality, Fibonacci numbers, and the Mersenne gcd identity used in our proofs. The Liquid Tensor Experiment [2] and the formalization of perfectoid spaces [1] demonstrate Lean 4's capacity for large-scale mathematical formalization. The Odd Order Theorem [5] and the Kepler Conjecture [6] are landmark formalizations in Coq and HOL Light, respectively.

Prime-producing formulas. The UGP prime formula (7) is a two-parameter family constrained by the Diophantine condition (6). This places it in the tradition of prime-producing polynomials (Euler's $n^2 + n + 41$, the Bunyakovsky conjecture) and prime-generating sieves, though the constraint to ridges $2^n - 16$ is novel. Exhaustive computation finds 57 terms up to 10^8.

Companion work. The UGP framework was introduced in [9]; the present formalization provides the machine-checked foundation. The self-reference theorems are imported from two companion Lean 4 libraries:

- **nems-lean** [10]: Lawvere fixed-point theorem in abstract form, Kleene recursion theorem, and Rogers' fixed-point theorem, all proved from a minimal self-referential interface (SRI).
- **aps-rice-lean** [8]: Rice's theorem and halting undecidability for acceptable programming systems, proved from the s-m-n theorem and a diagonal argument.

The companion **primes-in-greedy-B3** [11] provides a related B_3 prime result: every prime appears in the greedy multiplicative B_3 sequence (OEIS A079852).

17 Conclusion

We have presented ugp-lean, a self-contained Lean 4 formalization of the Universal Generative Principle. The artifact defines the UGP framework from first principles—ridge sieve, prime-

lock criterion, RSUC, GTE orbit, Quarter-Lock, Turing universality—with a strict zero-sorry, zero-custom-axiom policy on the core proof path.

The formalization contributes several results not present in prior work: the orbit-from-T derivation (theorem 6.1), the general ridge remainder lock (theorem 7.1), the even-step c-invariance theorem (theorem 6.2), the UGP prime class with ten machine-checked instances (theorem 8.3), the divisor count unboundedness theorem (theorem 11.3), the exact divisor-count formula $\tau(2^n - 16) = 5 \cdot \tau(2^{n-4} - 1)$ (theorem 11.4), the precise formal statement of the mirror-dual conjecture (theorem 11.5) with five machine-verified witness pairs, the mirror shift algebra (section 12): the universal law $c_1(b_2, q_2) - c_1(q_2, b_2) = b_1(q_2 - b_2)$ and the shared residue $c_1 \equiv 20 \pmod{b_1}$ for all mirror-dual pairs at all ridge levels, and a suite of uniqueness certificates (section 14.1): the derivation of $2b_1 - a_2 = 137$ from orbit data, non-periodicity of the canonical orbit, the symmetry group $\mathbb{Z}/2\mathbb{Z}$ of the mirror fiber, MDL canonical seeds at $n = 10, 13, 16$, order classification, and a certified singular series lower bound $\geq 85/100$.

What remains open. Six conjectures are formally stated in table 2. The mirror-dual conjecture (theorem 15.1) and UGP prime infinitude (theorem 15.2) are the most mathematically significant: both are structurally analogous to the twin prime conjecture and would require new sieve-theoretic techniques. The μ-flip conjectures concern the distribution of squarefree integers in arithmetic progressions and are open as formalization targets even where the mathematics is known.

Reproducibility. The artifact is publicly available at https://github.com/novaspivack/ugp-lean. Building requires LEAN 4 v4.29.0-rc6 and Mathlib v4.29.0-rc6. The command `lake build` compiles the entire library; no external tools beyond LEAN 4 and its package manager are needed.

Table 3: Complete theorem inventory. "Novel" indicates results new to this formalization; "General" indicates level-independent results. All theorems have zero `sorry` on the core proof path.

Theorem / Lean name	**Module**	**Paper ref**	**Novel?**
Core Classification (§5)			
`theoremA_general`	Classification.TheoremA	Thm A	—
`theoremB`	Classification.TheoremB	Thm B	—
`rsuc_theorem`	Classification.RSUC	RSUC	—
`mdl_selects_LeptonSeed`	Classification.TheoremB	MDL	—
`rsuc_formal`	Classification.FormalRSUC	RSUC	—
`rsuc_canon`	Classification.FormalRSUC	RSUC	—
`strengthening_cannot_add`	Classification.Monotonic	—	—
Ridge & Sieve (§5)			
`ridgeSurvivors_10`	Compute.Sieve	§5	—
`ridge_remainder_lock`	Core.RidgeRigidity	Lem m_2	—
`exclude_16` … `exclude_63`	Compute.ExclusionFilters	§5	—
GTE Orbit (§6)			
`orbit_determined_by_T`	GTE.UpdateMap	Prop 5.1	**Yes**
`update_map_produces_orbit`	GTE.UpdateMap	Prop 5.1	**Yes**
`even_step_c_invariance`	GTE.UpdateMap	—	**Yes**
`c3_strict_eq_c2_at_n10`	GTE.UpdateMap	—	**Yes**
`gen1_mersenne_isolation`	GTE.UpdateMap	—	**Yes**
`mersenne_entanglement_gen`	GTE.UpdateMap	—	**Yes**
Dynamics Companions / ML-9 (§6)			

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(Table 3 continued)

Theorem / Lean name	Module	Paper ref	Novel?
`gteTransition`	GTE.GTESimulation	—	**Yes**
`gteAfterSteps`	GTE.GTESimulation	—	**Yes**
`gte_s1...gte_s8`	GTE.GTESimulation	—	**Yes**
`gte_entropy_prefix8_gt_prefix9`	GTE.EntropyNonMonotone	ML-9	**Yes**
`Hpred8_gt_Hpred9`	GTE.EntropyNonMonotone	ML-9	**Yes**
General-Level Theorems (§7)			
`ridge_remainder_lock_gen`	GTE.UpdateMap	Lem m_2	**General**
`quotient_gap_all`	GTE.GeneralTheorems	gap-13	**General**
`mirror_b1_invariance`	GTE.UpdateMap	—	**Yes**
`mersenne_gcd_identity`	GTE.UpdateMap	—	Mathlib
`alpha_echo`	GTE.GeneralTheorems	α-echo	**Checked**
`b_linear_growth`	GTE.GeneralTheorems	b-growth	—
`c_values_strictly_mono`	GTE.GeneralTheorems	c-mono	—
Quarter-Lock & Kernel (§9)			
`quarterLockLaw`	QuarterLock	§3	—
`quarterLockStability`	QuarterLock	§3	—
`k_L2_eq`	ElegantKernel	§3	—
`L_model_from_gauge_struct`	LModelDerivation	§3	—
Universality & Self-Reference (§10)			
`uwca_P3_eq_rule110`	Universality.UWCASimul	Thm 10.1	**Yes**
`uwca_sweep_implements_rule110`	Universality.UWCASimul	Thm 10.1	**Yes**
`uwca_sector_invariant`	Universality.UWCASimul	Thm 10.1	**Yes**
`uwca_tile_verification`	Universality.UWCASimul	Thm 10.1	**Yes**
`uwca_simulates_rule110_real`	Universality.UWCAEmbeds	Thm 10.1	**Yes**
`ugp_is_turing_universal`	Universality.TuringUniv	Thm 10.2	—
`uwca_augmented_left_inverse`	Univ.UWCAHistoryReversible	T2	**Yes**
`uwca_history_lane_step_reversible`	Univ.UWCAHistoryReversible	T2	**Yes**
`ugp_lawvere_fixed_point`	SelfRef.LawvereKleene	Lawvere	Import
`ugp_kleene_recursion_thm`	SelfRef.LawvereKleene	Kleene	Import
`ugp_rice_theorem`	SelfRef.RiceHalting	Rice	Import
`ugp_halting_undecidable`	SelfRef.RiceHalting	Halting	Import
UGP Primes (§8)			
`IsUGPPrime`	GTE.UGPPrimes	Def 8.1	**Yes**
`ugp_primes_exist`	GTE.UGPPrimes	—	**Yes**
`ten_ugp_primes`	GTE.UGPPrimes	—	**Yes**
`ugp_primes_at_three_lvls`	GTE.UGPPrimes	—	**Yes**
`c1Val_eq_c1FromPair`	GTE.UGPPrimes	—	**Yes**
`UGPPrimeInfinitudeConj`	GTE.UGPPrimes	—	**Yes**
Mirror-Dual Conjecture (§11)			
`card_divisors_ridge_unb`	GTE.MirrorDualConj	—	**Yes**
`tau_ridge_exact`	GTE.MirrorDualConj	Thm 11.4	**Yes**
`coprime_pow2_mersenne`	GTE.MirrorDualConj	—	**Yes**
`five_mirror_dual_pairs`	GTE.MirrorDualConj	—	**Yes**
`MirrorDualConjecture`	GTE.MirrorDualConj	—	**Yes**
Mirror Shift Algebra (§12)			
`lepton_mirror_shift`	Core.MirrorAlgebra	Thm 12.3	**Yes**
`mirror_shift_factored`	Core.MirrorAlgebra	Thm 12.3	**Yes**
`lepton_shared_residue`	Core.MirrorAlgebra	Thm 12.1	**Yes**
`c1Val_mod_b1`	GTE.MirrorShift	Thm 12.1	**General**
`mirror_pair_shared_residue`	GTE.MirrorShift	Thm 12.1	**General**
`mirror_shift_formula`	GTE.MirrorShift	Thm 12.2	**General**
`mirror_shift_divides_b1`	GTE.MirrorShift	Thm 12.2	**General**

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(Table 3 continued)

Theorem / Lean name	Module	Paper ref	Novel?
lepton_shift_is_18_times_b1	GTE.MirrorShift	Thm 12.3	**Yes**
shift_n13	GTE.MirrorShift	Tab 1	**Yes**
shift_n16_a	GTE.MirrorShift	Tab 1	**Yes**
Companion modules (not in paper; see RESONANT_FACTORY_NOTE.md)			
qm_no_root_of_legendreSym_neg_one	GTE.InertPrimes	—	**General**
qm_inert_primes_certified	GTE.InertPrimes	—	**Yes**
kappa_unram_lower_bound	GTE.InertPrimes	—	**Yes**
discKminus, discKplus	GTE.ResonantFactory	—	**Yes**
rhoF_identity_verified	GTE.ResonantFactory	—	**Yes**
dickman_equidistribution_in_APs	GTE.AnalyticArch	—	Sorry (Tenenbaum)
qminus_qplus_coprime	GTE.AnalyticArch	—	**Yes**
Resonant Factory (§11.1)			
branch_linearization	GTE.ResonantFactory	Thm 11.6	**Yes**
factory_gap_two	GTE.ResonantFactory	Thm 11.8	**Yes**
factoryDp_prime	GTE.ResonantFactory	—	**Yes**
localDensity_3..43	GTE.ResonantFactory	Thm 11.9	**Yes**
hasse_check_no_obstruction	GTE.ResonantFactory	Thm 11.9	**Yes**
Structural Theorems (§14)			
golden_ratio_minimal_poly	GTE.StructuralThm	Thm 14.1	**Yes**
fib_eigenvalue_product	GTE.StructuralThm	Thm 14.1	**Yes**
fib_contraction_rate	GTE.StructuralThm	Thm 14.1	**Yes**
mirror_fiber_two	GTE.StructuralThm	Thm 14.2	**Yes**
mirror_pair_induces_loop	GTE.StructuralThm	Thm 14.2	**Yes**
minimality_duality_n10	GTE.StructuralThm	Thm 14.3	**Yes**
only_survivors_n10	Compute.ExclFilters	Thm 14.3	**Yes**
fingerprint_fixed_point_exists	GTE.StructuralThm	Thm 14.4	Sorry (Tarski)
decidability_phase_transition	GTE.StructuralThm	Thm 14.5	**Yes**
leptonSeed_is_lex_min_residual	GTE.StructuralThm	—	**Yes**
Uniqueness Certificates (§14.1)			
derivation_of_137	GTE.UniquenessCert	Thm 14.6	**Yes**
derivation_of_137_structural	GTE.UniquenessCert	Thm 14.6	**Yes**
alpha_echo_per_level	GTE.UniquenessCert	Thm 14.6	**Yes**
orbit_not_eventually_periodic	GTE.UniquenessCert	Thm 14.7	**Yes**
rule110_tiles_are_minimal	GTE.UniquenessCert	Thm 10.3	**Yes**
mirror_symmetry_group_is_Z2	GTE.UniquenessCert	Thm 10.4	**Yes**
no_order_3_loop	GTE.UniquenessCert	Thm 10.4	**Yes**
mdl_c1_n13_is_9007	GTE.UniquenessCert	Thm 14.8	**Yes**
mdl_c1_n16_is_46681	GTE.UniquenessCert	Thm 14.8	**Yes**
order_classification_n10_to_n13	GTE.UniquenessCert	Thm 14.9	**Yes**
singular_series_partial_lower_bound	GTE.UniquenessCert	Thm 14.10	**Yes**
n10_uniqueness_package	GTE.UniquenessCert	§14.1	**Yes**
AsymptoticSparsityConjecture	GTE.UniquenessCert	Conj 15.5	Stated
UGPZetaFactorizationConjecture	GTE.UniquenessCert	Conj 15.6	Stated
Additional proved results (from original conjectures)			
mirror_min_dual_proved	Conjectures	Thm 7.3	**Yes**
fib_rigidity_proved	Conjectures	Thm 7.2	**Yes**
c1_monotone_in_q2	Conjectures	Thm 7.4	**Yes**
robust_universality_proved	Conjectures	Thm 10.2	**Yes**
sharp_boundary_proved	Conjectures	§10	**Yes**

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(Table 3 continued)

Theorem / Lean name	Module	Paper ref	Novel?
`kernel_compatibility_proved`	Conjectures	Thm 9.1	**Yes**
`global_c_attractor_proved`	Conjectures	Thm 6.2	**Yes**
Real Analysis / DSI Export (§13)			
`ugpOutputGap_deriv`	GTE.DSIExport	Thm 13.2	**Yes**
`ugp_deriv_lower_bound`	GTE.DSIExport	Thm 13.3	**Yes**
`ugpOutputGap_differentiableOn`	GTE.DSIExport	—	**Yes**

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